

Interview Q&A: IImS

1. What is the difference between a static and dynamic memory allocation in LLMS?

In LLMS, static memory allocation involves reserving memory at compile time for variables or data structures that are known to have a fixed size, whereas dynamic memory allocation involves requesting memory from the system at runtime using functions like `malloc()` or `new`.

2. How does the Longest Increasing Subsequence (LIS) problem relate to dynamic programming?

The LIS problem is a classic example of dynamic programming, where we need to find the length of the longest increasing subsequence in an array.

3. What is the time complexity of the Longest Increasing Subsequence (LIS) problem using dynamic programming?

The time complexity of LIS using dynamic programming is $O(n^2)$, where n is the number of elements in the input array.

4. What is the time complexity of the Longest Increasing Subsequence (LIS) problem using dynamic programming with a bottom-up approach?

The time complexity of the LIS problem using dynamic programming with a bottom-up approach is $O(n^2)$, where n is the length of the input sequence.

5. What is the time complexity of the Longest Increasing Subsequence (LIS) problem using dynamic programming with a top-down approach?

The time complexity of the LIS problem using dynamic programming with a top-down approach is $O(n*m)$, where n is the length of the input sequence and m is the maximum value in the sequence.

6. What is the key difference between memoization and tabulation in dynamic programming?

Memoization stores the results of expensive function calls and reuses them when the same inputs occur again, whereas tabulation fills up a table with the solutions to subproblems before solving the main problem.